

How to Rocketry

Link to workshops from previous FIRST seasons:

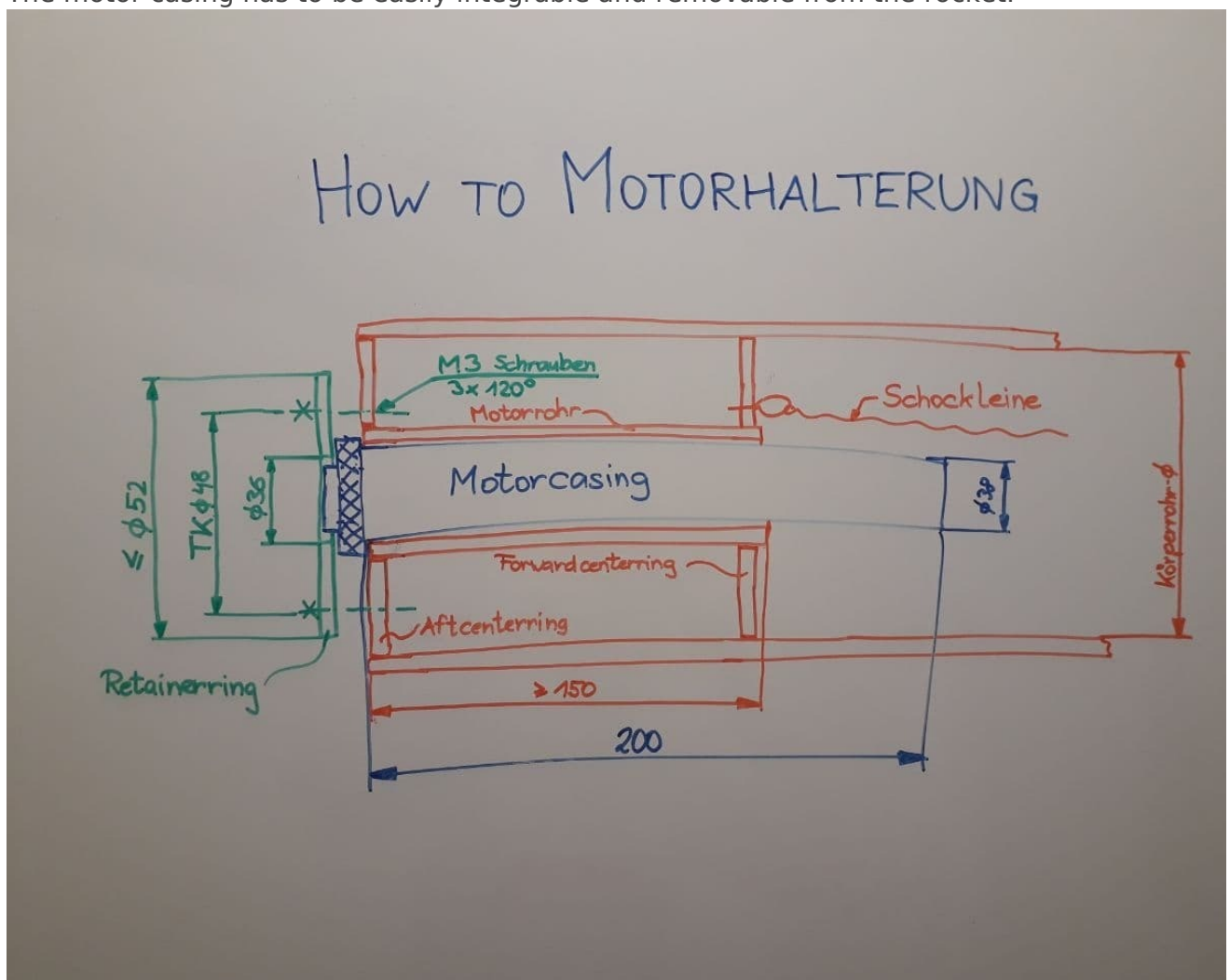
https://youtube.com/playlist?list=PLF_ij5FHIMIT2lomy2YwLGfMxNMTDnl-G

¹Fins

- 1.1 Use fillets at edges between bodytube and fins
- 1.2 When slotting fins into the bodytube, do not cut the bodytube all the way to the end - this leads to structural weaknesses
- 1.3 Be careful when fins protrude behind the bodytube - this can cause fins to break upon ground impact. Fins can be designed as replaceable to avoid this issue.
- 1.4 Aerodynamic stability can be increased with spin, although it's not very usual due to the increased manufacturing complexity.

²Motor mount

- 2.1 The motor casing has to be easily integrable and removable from the rocket.



2.2 The motor casing has to be held inside the motor tube, usually with a retainer ring or a threaded tailcone.

2.3 Make sure that no screws or retainer parts protrude unnecessarily far behind the body tube.

2.4 Nomenklatur:

2.4.1 Grain: solid fuel/oxidator mixture, that gets burned and produces the thrust

2.4.2 Reload: consists of grains and other single use parts, that can be bought as a package (e.g. Aerotech H2389T)

2.4.3 Motorcasing: reusable metal casing that holds the grain - it's classified by their diameter and length in millimeters (e.g. 29/180)

2.4.4 Motorrohr/Motorhalter: tube (often cardboard) that is fixed inside the back of the rocket, in which the motor casing can be inserted and fixed

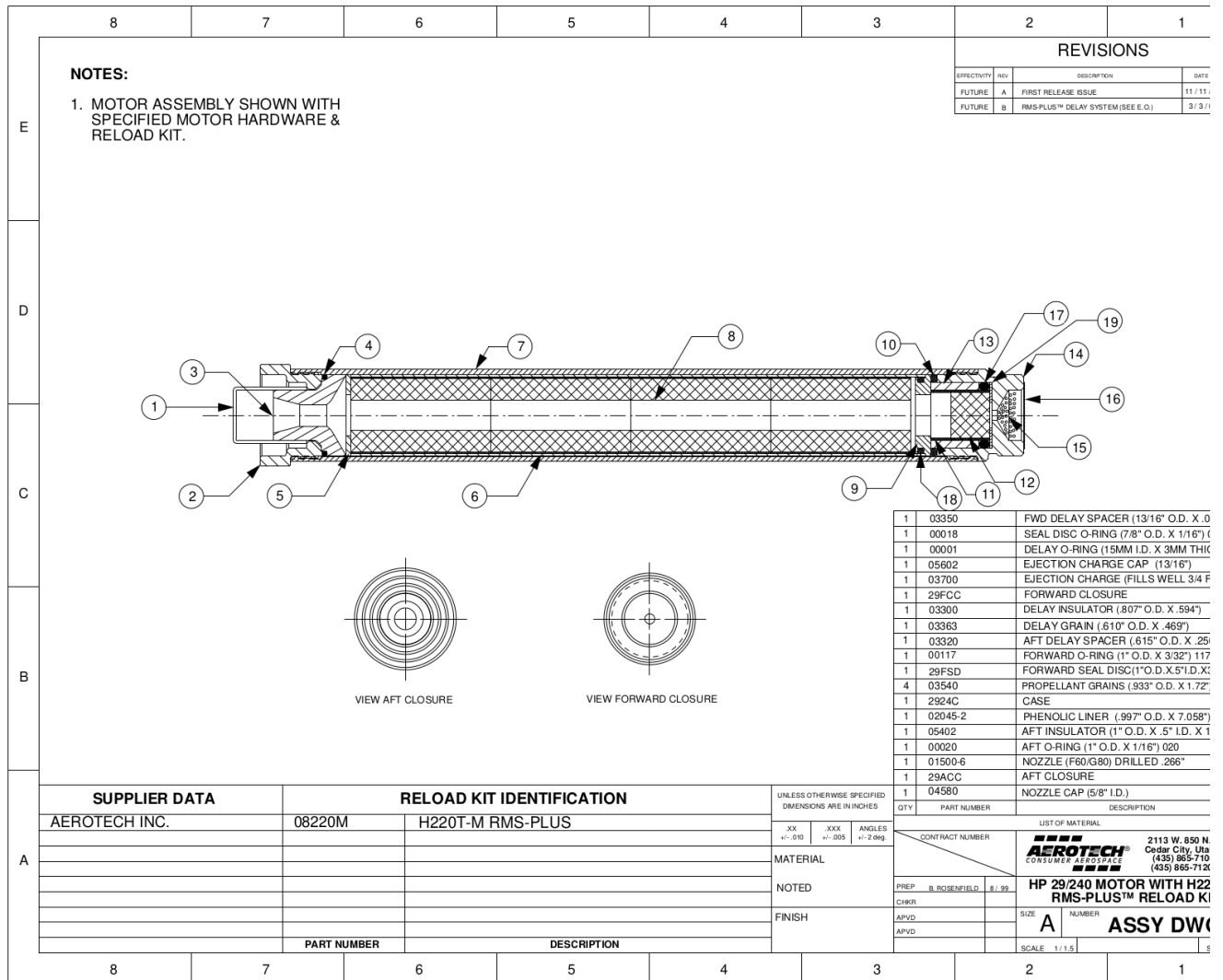
2.4.5 Adapter: can be used to put smaller motor casings into motor mounts with a larger diameter, e.g. using a 29 mm motor casing in a 38 mm motor mount

2.4.6 Retainer: hold the motor casing inside the motor tube

2.4.7 Spacer: allows using casings, that would normally require a more powerful motor, with a smaller one by replacing grains with spacers

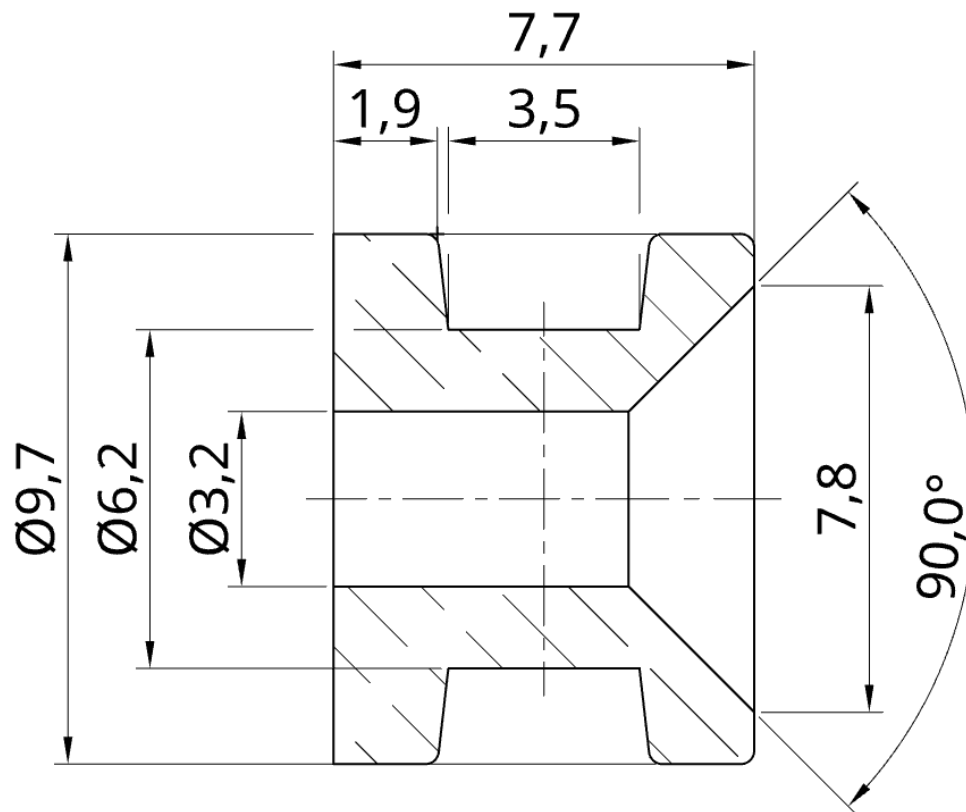
3 Motor

- 3.1 We usually use a Aerotech H-238T with a RMS-29/180 casing or a RMS-29/240 casing with one spacer. The image shows a section cut of an Aerotech H220T motor.



4Railbuttons

- 4.1 one should be placed far back on the rocket, so that the rockets remains guided in the launch rail for as long as possible, while the other one should be located around the rocket's center of mass
- 4.2 if the railbuttons are placed on seperate body tube segments, they should be designed to prevent rotation between the two bodytube segments
- 4.3 the mount of the railbuttons on the inside of the rocket must not obstruct the parachute from deploying
- 4.4 measurements of the railbutton:



5 Flight computer

- 5.1 The flight computer detects the apogee (highest point of the flight) and ignites a black powder charge inside the bodytube. The created pressure forces the rocket segments to separate, allowing the parachute to slip out of the bodytube.
- 5.2 The height measurements works with a barometric sensor, therefore the flight computer requires some holes (at least two 2 mm holes at oppositbe sides of the bodytube) to allow it to detect the outside pressure. These holes have to be radial to the rocket's axis, so that the sensor only measure static pressure and not dynamic pressure from the "wind" of the air in the rocket's path.
- 5.3 The flight computer has to be protected from the hot gas created by the blackpowder.
- 5.4 muss im zusammengebauten Zustand auf der Startrampe scharf gemacht und entschärft werden können (RBF Pin, Schalter...) The flight computer has to be able to be armed and disarmed after the rocket has been inserted into the launch rail with the use of a switch or RBF (remove before flight) pin.

6 Radio transmission

- 6.1 If the rocket has to communicate with the ground in any way, be aware that carbon fiber is conductive - use glass fiber for such segments instead.

7 Parachute

- 7.1 Has to easily slip out of the rocket - if the rocket is turned upside down, it should slip out without require any shaking.

- 7.2 Use slim Kevlar lines to connect the parachute, they are very heat resistant to survive the hot black powder gases.
- 7.3 If the parachute main line should be removable, fix shorter lines into the rocket where the parachute main line can be connected with a carabiner.
- 7.4 Leine am Körperrohrende wird üblicherweise am Centerring der Motorhalterung fixiert. Sie ist eher lang (~3-5 m), damit die Rakete nicht zu sehr taumelt und sich die Leine beim Öffnen des Fallschirms mehr dehnen kann. Die plötzliche Öffnung erzeugt nämlich einen Schock, der das Seil, den Ankerpunkt oder das Körperrohr beschädigen können. An der Stelle, an der die Leine das Körperrohr bei der Trennstelle berührt, kann mit Kreppband verdickt werden, damit die Kraft der Leine verteilt wird. Das Kreppband kommt auf die Leine, nicht das Rohr. The parachute's main line is usually fixed on one of the motor tubes centering rings and is about 3-5 m long. To avoid the parachute line cutting into the bodytube due to the shock of the parachute deployment, this section of the bodytube can be supported with crepe tape.
- 7.5 The parachute is not exerted by the blackpowder charge but due to the relative speed between the rocket's segments.
- 7.6 The parachute line of the nosecone can be shorter (~1 m).
- 7.7 The rocket also has some drag when descending, which is not considered by dimensioning software. Therefore the actual parachute may be slightly smaller than the calculated size.
- 7.8 Ideally, the rocket should be aerodynamically instable after separation, leading it to tumble and the parachute to deploy more easily.

8 **Seperation point**

- 8.1 The coupler's shoulder length (i.e. length inside the rocket part that gets removed at separation) has to be at least one caliber (caliber = bodytube diameter)!!!
- 8.2 Test for the friction coupler: when holding the lighter section of the rocket, the rocket should not separate. With some shaking, the friction coupler should loosen and separate. The strength of the friction coupler can be adjusted with crepe tape, for example.
- 8.3 Be aware that the strength of the friction coupler can change due to environmental influence. For example, on a cold launch day, the friction coupler will be looser.

9 **Blackpowder charge**

- 9.1 small plastic capsule with an e-match, usually lies in the rocket without fixation
- 9.2 has to be in the same pressure hull as the motor's back delay charge
- 9.3 has to be wired to the flight computer
- 9.4 has to be separated from the flight computer to not cause damage

10 **to speed up the preparations and reduce complications it is advised to**

- 10.1 make the flight computer and battery easily replaceable
- 10.2 allow an easy integration and removal of the motor
- 10.3 make the parachute and parachute lines easily replaceable

11 **other tips**

11.1 stick to one type of screw (e.g. Torx M3)

11.2 check out the safety code of the Tripoli (renowned association for high-power model rocketry) - not mandatory but may be interesting: <https://www.tripoli.org/safety>

Common materials

1 Screws, thread inserts

2 epoxy, superglue

3 wooden plates, glass fiber (GFK) plates

4 PLA, PC, ASA for 3D printing

5 Aluminium (tube, cylinder, plate)

6 GFK (carbon fiber)/CFK-prepreg for laminating

7 cables

Manufacturing methods

1 laminating (wet, prepreg)

2 CNC milling (wood, GFK...)

3 lathing (metals)

4 3D-printing (PLA, PC, ASA)

5 "Dremel", drilling (Danger: CFK, GFK-dust is toxic)

6 Glueing (epoxy, super glue)

7 sewing (sewing machine)

8 splicing (parachute lines)

9 spray paint

10 foiling

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